
Software Requirements Specification

for

<KADA System>

Version 1.0 approved

**Prepared by
Cheryl Cheong Kah Voon
Chau Ying Jia
Lau Yee Wen
Neo Li Xin
Woo Cheng Shuan**

**Faculty of Computing,
University Technology Malaysia**

2 November 2024

Table of Contents

Table of Contents.....	ii
Revision History.....	ii
1. Introduction.....	1
1.1 Purpose.....	1
1.2 Document Conventions.....	1
1.3 Intended Audience and Reading Suggestions.....	1
1.4 Product Scope.....	1
1.5 References.....	1
2. Overall Description.....	2
2.1 Product Perspective.....	2
2.2 Product Functions.....	2
2.3 User Classes and Characteristics.....	2
2.4 Operating Environment.....	2
2.5 Design and Implementation Constraints.....	2
2.6 User Documentation.....	2
2.7 Assumptions and Dependencies.....	3
3. External Interface Requirements.....	3
3.1 User Interfaces.....	3
3.2 Hardware Interfaces.....	3
3.3 Software Interfaces.....	3
3.4 Communications Interfaces.....	3
4. System Features.....	4
4.1 System Feature 1.....	4
4.2 System Feature 2 (and so on).....	4
5. Other Nonfunctional Requirements.....	4
5.1 Performance Requirements.....	4
5.2 Safety Requirements.....	5
5.3 Security Requirements.....	5
5.4 Software Quality Attributes.....	5
5.5 Business Rules.....	5
6. Other Requirements.....	5
Appendix A: Glossary.....	5
Appendix B: Analysis Models.....	5
Appendix C: To Be Determined List.....	6

Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of this document is to give a detailed description of the requirements for the “KADA system”. It will illustrate the purpose and complete declaration for the development of the system. It will also explain system constraints, interface and interactions with other external applications. This document is primarily intended to be proposed to a customer for its approval and as a reference for developing the first version of the system for the development team.

1.2 Document Conventions

Fonts	Chapter Names: 18 pt Times New Roman font in Bold Headings: 14 pt Times New Roman font in Bold Paragraph: 12 pt Times New Roman font, justify
Glossary	KADA - Koperasi Kakitangan Daerah Kelantan Berhad

1.3 Intended Audience and Reading Suggestions

The project is a prototype for the KADA system and it is restricted to use within the organization. It has been designed, developed and implemented under the guidance of KADA’s representative. This project is useful for both the KADA management team and the members by improving some processes like member registration and loan application process.

1.4 Product Scope

The KADA system is a cloud-based platform with integrated AI, designed to automate and streamline KADA’s internal processes, particularly member registration and loan application. This system aims to reduce manual work, improve operational efficiency and enhance usability through a digital solution. KADA system provides a combined platform for administrators, members and the board of directors with the following core functionalities such as member management, loan application and processing, data analytics and reporting.

1. Member Registration

The new user can register their membership by submitting the member application online to reduce the burden of manual data handling and processing.

2. Loan Application

The loan application can also submitted online through the system.

3. Application status checking

This feature can allow the applicant to check the real-time status of their application through the website. Once the application is approved, they will receive an automated email or SMS notification.

4. Check financial status and statement

The system will

5. Check monthly and annual report

The system will generate monthly and annual reports for the members from the database to allow members can know their financial performance from time to time. The system will also generate an annual report for the board of directors to offer a comprehensive overview

of the financial indicators and operational trends to support the strategy's decision-making.

1.5 References

<List any other documents or Web addresses to which this SRS refers. These may include user interface style guides, contracts, standards, system requirements specifications, use case documents, or a vision and scope document. Provide enough information so that the reader can access a copy of each reference, including title, author, version number, date, and source or location.>

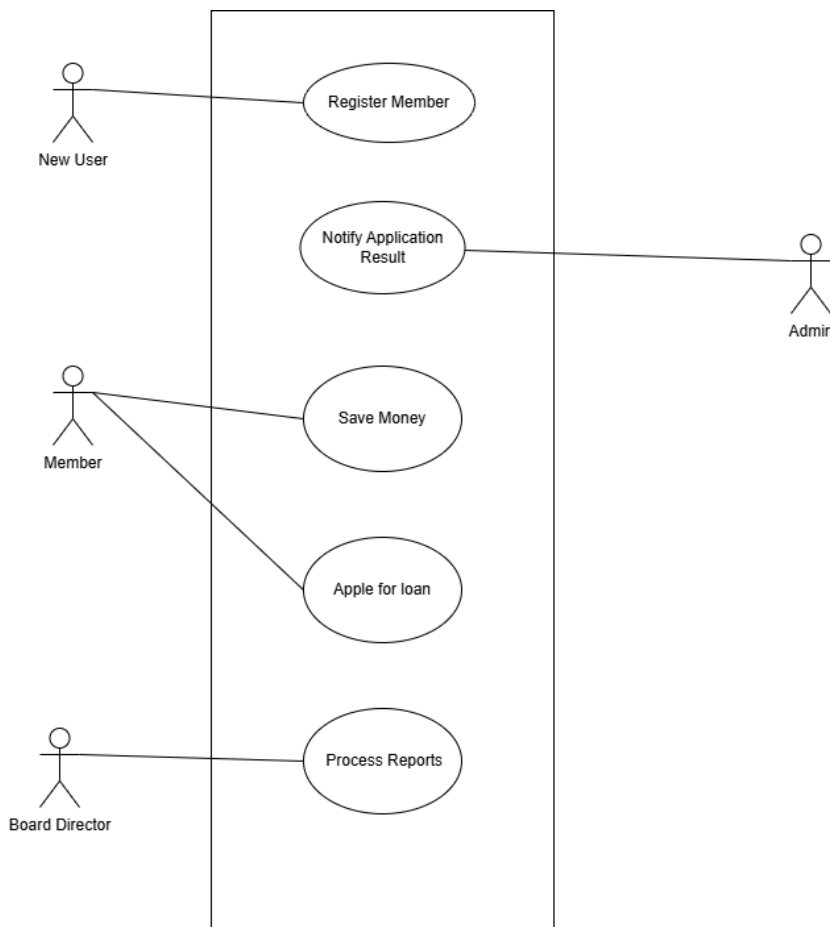
1. <https://krazytech.com/projects>

2. Overall Description

This section will provide a comprehensive overview of the entire system. It will explain the system's context, illustrating how it interacts with other systems while introducing its fundamental functionalities. Additionally, it will identify the various stakeholders who will utilize the system and outline the specific functionalities available to each group. Finally, the constraints and assumptions related to the system will be discussed.

2.1 Product Perspective

The KADA Registration System is an independent platform specifically created to streamline manual processes at Koperasi Kakitangan KADA Kelantan Berhad. This system consolidates multiple functions, such as member registration, application processing, savings management, and loan application processing, into one digital solution. It connects to a secure database for storing and managing user and financial information, while also integrating with email and SMS services to keep users informed about updates. Utilizing role-based access control, it differentiates functionalities for users, administrative staff, and board directors. Furthermore, it aims to improve operational efficiency by automating tasks that once required manual effort.



2.2 Product Functions

The KADA Registration System serves several key functions, including:

Member Registration: Facilitates new users in submitting their personal information for registration while tracking their application status.

Savings Management: Empower members to deposit and manage their savings, access financial statements, and monitor dividends.

Loan Applications: Offers a streamlined process for members to submit loan applications, complete with built-in eligibility checks.

Application Processing: Enables administrators to review, approve, or decline applications and loan requests.

Notifications: Sends automated updates to members via email or SMS regarding application progress and financial information.

Report Generation: Produces reports for board directors, summarizing application statuses, loan approvals, and other essential data for effective oversight and decision-making.

2.3 User Classes and Characteristics

The system categorizes users into the following classes:

New Users:

- Individuals who have not yet registered and are interested in becoming members. They are generally expected to have a basic understanding of digital forms.

Members:

- Registered users who can apply for loans, manage savings, and monitor financial updates. Their levels of digital literacy may vary.

Administrative Staff (Admins):

- KADA employees tasked with verifying member and loan applications. They need a moderate level of familiarity with the system to ensure efficient processing.

Board Directors:

- Senior decision-makers overseeing loan approvals and high-level reporting. They primarily need system access for review and approval purposes.

2.4 Operating Environment

The system will be a web-based application accessible via modern web browsers (e.g., Google Chrome, Mozilla Firefox) on desktops, laptops, and mobile devices. It will use HTTPS protocols for secure communication. The backend will operate on a secure, cloud-based infrastructure that supports scalability, data encryption, and backups.

2.5 Design and Implementation Constraints

1. **Security Compliance:** The system must adhere to data protection regulations, such as GDPR, and implement Two-Factor Authentication (2FA) to safeguard member information.
2. **Data Integrity:** It is essential to ensure the precise processing and storage of sensitive data, with redundancy measures established.
3. **Scalability:** The system should accommodate a growing user base, necessitating a scalable infrastructure.
4. **Cross-Browser Compatibility:** The system must function seamlessly across multiple web browsers and devices.
5. **Role-Based Access Control:** The design of the system must enforce distinct permissions and access controls tailored to each user class.

2.6 User Documentation

1. **User Manuals:** Manuals for Members, Admins, and Board Directors, covering system functionalities and tasks specific to each user class.
2. **Online Help System:** In-system help features, including tooltips, FAQs, and context-sensitive guidance.
3. **Interactive Tutorials:** Onboarding and feature-specific tutorials to guide users through key functions.
4. **Knowledge Base and FAQs:** A searchable online resource with common questions, troubleshooting tips, and best practices.
5. **Video Tutorials:** Short instructional videos for each user class, segmented by topic.
6. **Documentation Delivery Formats:** Available as PDF files for offline use and HTML for in-system access, following accessibility and usability standards.

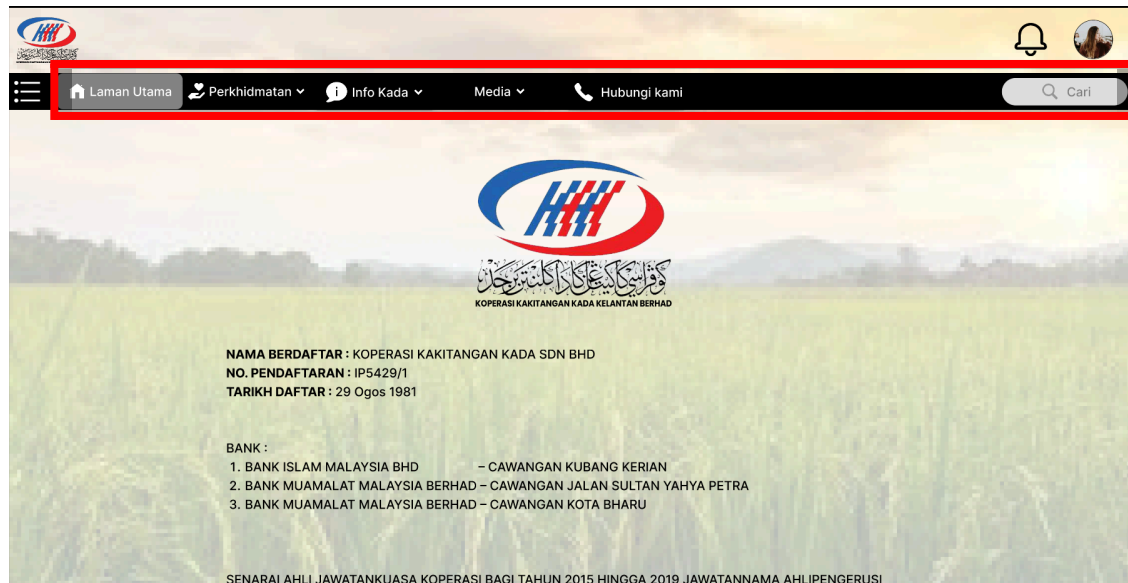
2.7 Assumptions and Dependencies

- Users can connect to the system through the internet using compatible devices.
- Administrative staff and board directors are well-versed in digital systems and need little training.
- The system depends on dependable email and SMS gateways to send notifications.
- Key dependencies include the security standards and cloud infrastructure providers that facilitate database and application hosting.

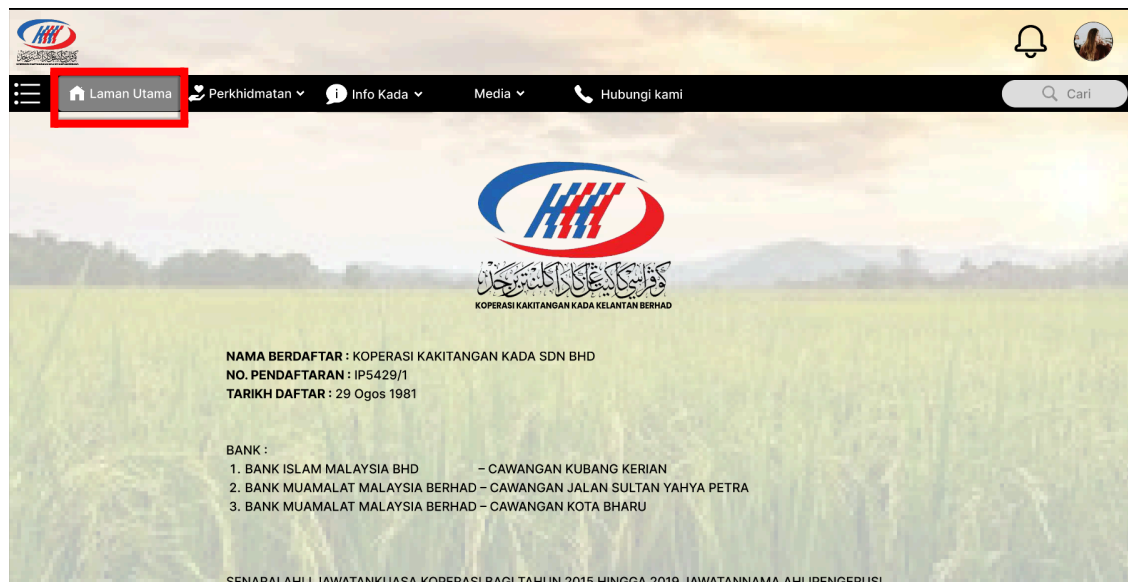
3. External Interface Requirements

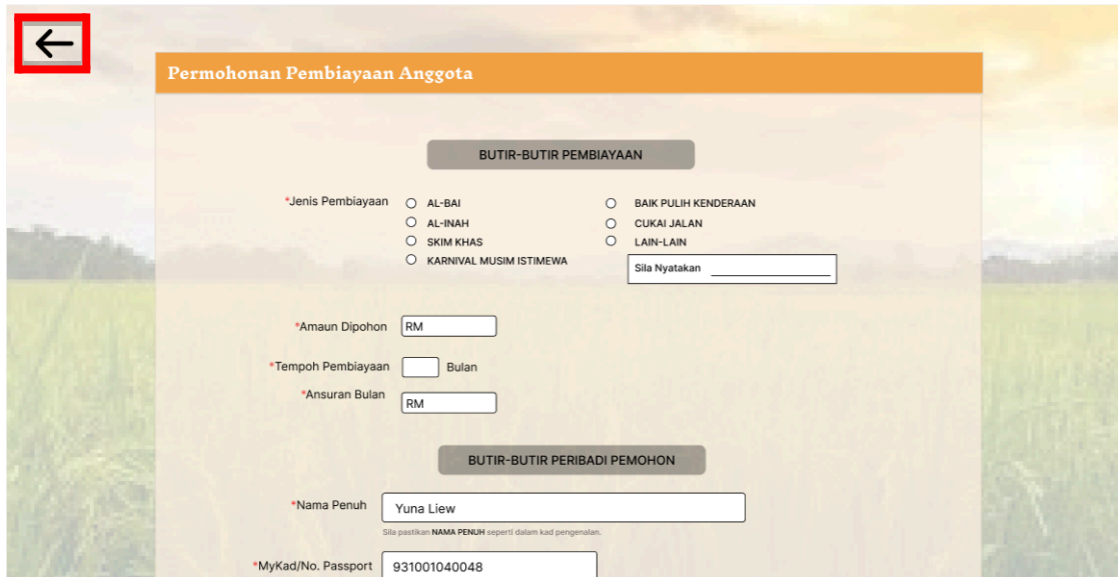
3.1 User Interfaces

Most of the functions are placed in the top panel which includes the homepage, services, informatics, media, contact information and search. This is to improve the visibility and accessibility of these functions.



Every page of the user interface is equipped with a “home” button or a “back” button. The “back” button allows the user to return to the previous page while the “home” button sends the user back to the homepage. These buttons are highly essential to allow users to correct their mistakes and provide comfort in navigation. These buttons are designed to be placed on the top panel as users are more accustomed to finding them there.





Permohonan Pembiayaan Anggota

BUTIR-BUTIR PEMBIAYAAN

*Jenis Pembiayaan

☐ AL-BAI ☐ BAIK PULIH KENDERAAN
☐ AL-INAH ☐ CUKAI JALAN
☐ SKIM KHAS ☐ LAIN-LAIN
☐ KARNIVAL MUSIM ISTIMEWA

Sila Nyatakan _____

*Amaun Dipohon RM _____

*Tempoh Pembiayaan _____ Bulan

*Ansuran Bulan RM _____

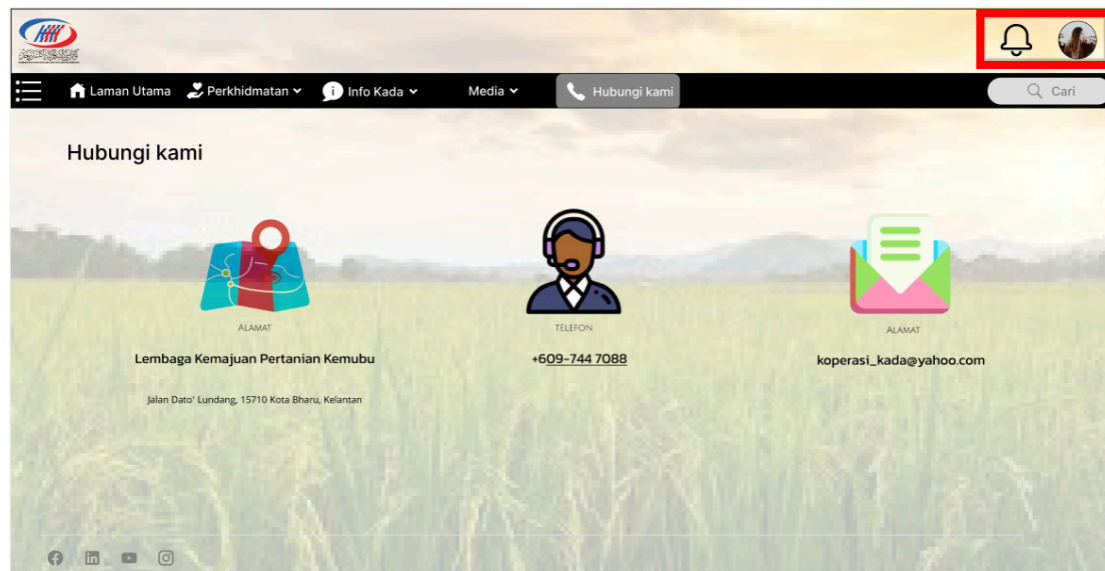
BUTIR-BUTIR PERIBADI PEMOHON

*Nama Penuh Yuna Liew

Sila pastikan NAMA PENUH seperti dalam kad pengenalan.

*MyKad/No. Passport 931001040048

Besides, the profile icon is placed on the top right so users can update their personal information and check the status of the application process anytime. For convenience, a notification bell is also added beside the profile to inform the user about any ongoing activity. These buttons are placed on the top right of the user interface for better navigation purposes.



Yuna Liew

Profil

Pinjaman

Permohonan

Daftar Keluar

Nama: Yuna Liew Mei Mei

Alamat Emel: Yuna Liew Mei Mei

MyKad/No. Passport: 000000-00-0000

Taraf perkahwinan: Kahwin

Alamat Rumah: 33, Jalan Bunga Tulip

Poskod: 00000

Negeri: Johor

Jantina: Perempuan

Agama: Buddhist

Yuna Liew

Profil

Pinjaman

Permohonan

Daftar Keluar

Status permohonan pembiayaan

1 Permohonan serahkan

2 Permohonan dinilai oleh pengarah lembaga

3 Permohonan lulus / gagal

4 Keputusan pemberitahuan

Pemberitahuan

✉ Permohonan pembiayaan lulus.

During the loan application, users are required to fill in their personal details in an online form. The answers can vary from multiple choices to text-based responses, based on the questions. After the completion of filling details, users are required to press the submit button to submit their application. The application will then be sent into a database system for further analysis and approval.

←

Permohonan Pembiayaan Anggota

*Jantina ☐ Lelaki ☒ Perempuan

*Agama

*Bangsa

*No. Anggota

*No. PF

*Jawatan& Gred

*Alamat Pejabat
(Tempat Bertugas)

←

Permohonan Pembiayaan Anggota

PENGESAHAN MAJIKAN

Kami mengesahkan bahawa : No. K/P :

telah memberikan butir-butir peribadi dan maklumat pendapatan selaras dengan rekod pekerjaan kakitangan tersebut.

Kami juga mengesahkan bahawa kakitangan adalah berjawatan tetap.

Gaji pokok sebulan

Gaji bersih sebulan



Tandatangan Sah
Bertarikh: 19/06/2024

Hantar

3.2 Hardware Interfaces

After the submission of a loan or member application, data from the user side will be transported to the server. The data transmitted is encrypted for better security. The data received by the server is stored in a hard disk drive (HDD) or storage device while the software manages read/write operations given by the user.

Supported devices include wired (Ethernet) and wireless network connection (Wi-Fi), providing flexibility in the network structure. For the communication protocols, Hypertext Transfer Protocol Secure (HTTPS) is used for data encryption, hence protecting the user's sensitive information during transfer.

3.3 Software Interfaces

The users' information is stored in the database such as MYSQL 8.0. The software also supports common operating systems such as Windows 11 or MacOS. The operating system manages hardware resources, file I/O and network functions, while the framework provides a foundation for managing HTTP requests. The software will send information such as users personal details, loan application and

payment status, while the server will send back information such as reports, email notification and database updates. Communication between components occurs primarily over RESTful APIs with structured JSON data, secured through protocols like HTTPS. Data that can be shared across software components are configurational data, user data, transaction data and graphical data.

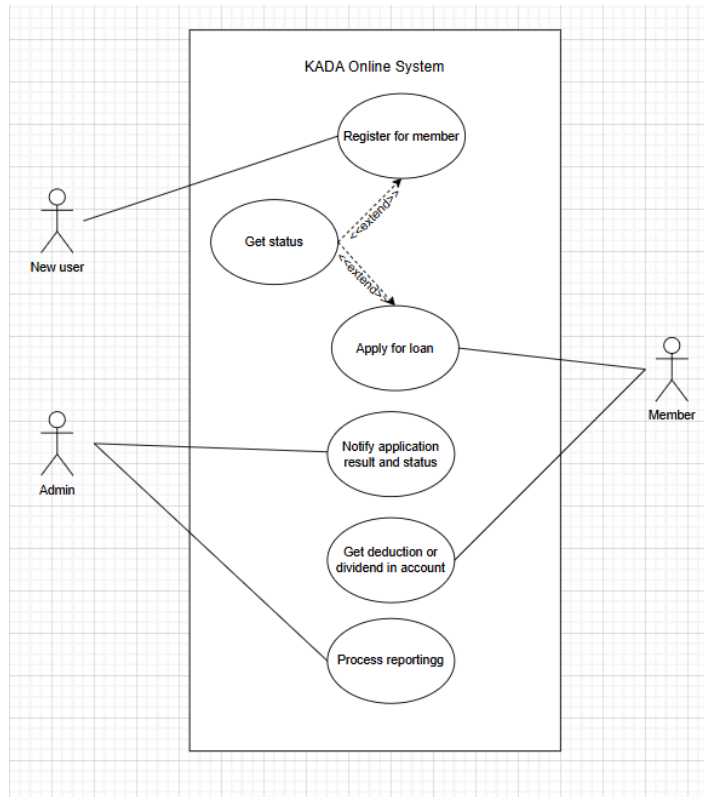
In order to ensure system requirements are met by the client, constraints need to be identified and documented. One of the constraints is that all shared data must be stored within the multitasking operating system so that it's accessible by multiple components. Besides, the global data area must be stored in shared memory, which brings up another memory management constraint. Lastly, to reduce chances of error, the global data area must implement robust error handling to manage corrupted data.

3.4 Communications Interfaces

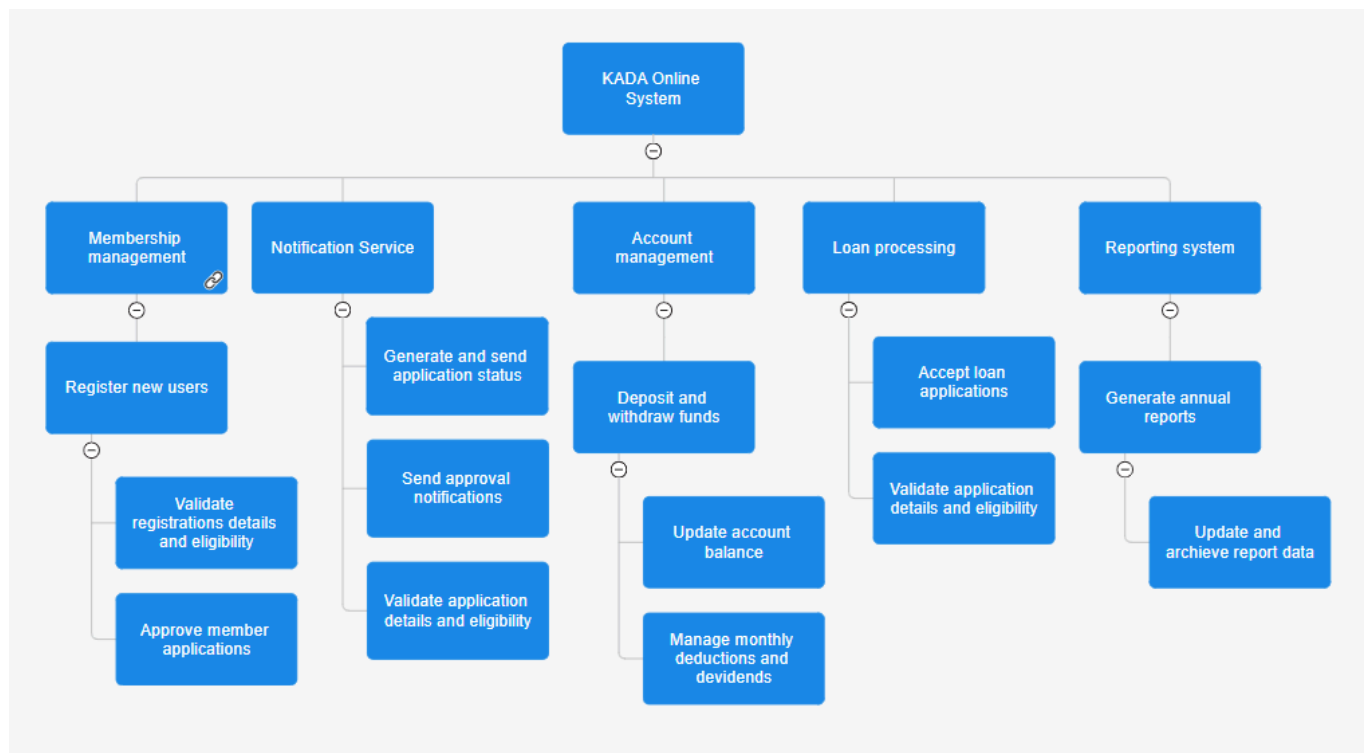
From the web browser aspect, the software can support common web browsers such as Google or Mozilla Firefox to access the application portal. The data submitted through the online form should follow JSON or XML format so the back-end system can process it. For the email communication aspect, it should be able to send instant notification to their email services, such as submission confirmation and application status. Other than that, the software also communicates with external banking systems or payment gateways.

For better networking processes, the software supports HTTPS for communication between user and the server, ensuring a safe data transferring process through data encryption. The software also uses HTTP/2.0 standard for stable network connection and faster data transmission rate.

4. System Features



Use case diagram



functional hierarchy

4.1 Member Registration

4.1.1 Description and Priority

Allow new users to register as members for loan applications and to get other benefits. Users should provide their personal information and other details through the website. It is categorized as a high-priority with a benefit rating of 9, a penalty rating of 7, a cost rating of 4 and a risk rating of 4.

4.1.2 Stimulus/Response Sequences

The new user enters their information and details for registration for example username, email and password. On the other hand, the system will verify and validate the input that the user enters. The system will also create a new user record for the board director to review and set the membership status to “Pending” in the user's account.

4.1.3 Functional Requirements

REQ-1: The system shall provide hints for users at specific fields for example the format for phone number and the identity card number.

REQ-2: The system shall validate that all required fields are fully filled out before submission.

REQ-3: The system shall save the user details and update the status of “Pending” in users account for further approval from admin.

REQ-4: The system shall send a confirmation message to the user's email after form submission.

4.2 Member Registration and Loan Application Approval

4.2.1 Description and Priority

Admin is responsible to review and approve or reject the new users' membership registration and members' loan application. The priority for this feature is high, with a benefit rating of 9, a penalty rating of 7, a cost rating of 3, and a risk rating of 5.

4.2.1 Stimulus/Response Sequences

Admin selects users registration and loan application to review whether they reach the requirements and are eligible. On the other hand, the system should retrieve the registration and loan application details for review and provide options for the admin to approve or reject.

4.2.3 Functional Requirements

REQ-1: The system shall display all pending registration and applications in the admin page for review.

REQ-2: The system shall update the registration and application status to “Approved” or “Rejected” based on the admin's decision.

REQ-3: The system shall record and save the approval or rejection details for board director documentation use.

4.3 Loan Application

4.3.1 Description and Priority

Allow members to apply for a loan through the KADA website. KADA will give a dividend annually for the members. The priority for this feature is high, with a benefit rating of 8, a penalty rating of 7, a cost rating of 5, and a risk rating of 7.

4.3.2 Stimulus/Response Sequences

Member should complete the loan application form, fill in the details and submit for application. While the system should validate the input and save the members' application records for board director. The system will then set the loan application status to "Pending" in the member's account.

4.3.3 Functional Requirements

REQ-1: The system shall provide hints for users at specific fields then have the format to be followed.

REQ-2: The system shall require that all mandatory fields are filled before submission.

REQ-2: The system shall mark the application status as "Pending" for the admin to review and approve or reject.

REQ-3: The system shall notify the member via email that the application was received.

4.4 Generate and Send Application Status

4.3.1 Description and Priority

Send notifications and emails to members or new users about the registration and application approvals or rejections. The priority for this feature is medium, with a benefit rating of 7, a penalty rating of 6, a cost rating of 3, and a risk rating of 3.

4.3.2 Stimulus/Response Sequences

The trigger is when an application status is being updated by the admin whether it is approved or rejected. The system will then be stimulated to generate a notification and status update and send it to the member or new user.

4.3.3 Functional Requirements

REQ-1: The system shall automatically detect the changes in application status and update on time.

REQ-2: The system shall also send a notification to the applicants and members' registered email address to know the status update.

4.4 Manage Monthly Deductions and Dividends

4.4.1 Description and Priority

Automatically manages monthly salary deductions and annual dividend credits for each member. The priority for this feature is high, with a benefit rating of 8, a penalty rating of 6, a cost rating of 6, and a risk rating of 5.

4.4.2 Stimulus/Response Sequences

System initiates monthly deductions and dividend calculation based on the scheduled dates. The system will then be stimulated to deduct a set percentage from each member's salary and add dividends to their savings.

4.4.3 Functional Requirements

REQ-1: The system shall calculate and deduct the specified percentage from each member's salary monthly.

REQ-2: The system shall calculate dividends based on the annual savings balance and credit members annually.

REQ-3: The system shall notify members of deductions and dividend credits via email.

4.5 Reporting System

4.4.1 Description and Priority

Compiles data on all user and member records, activities, generating an annual report for board review. The priority for this feature is medium, with a benefit rating of 9, a penalty rating of 4, a cost rating of 5, and a risk rating of 5.

4.4.2 Stimulus/Response Sequences

System will generate the records saved every time there is a new registration or loan application. The system will gather all the data and compile it into a comprehensive annual report.

4.4.3 Functional Requirements

REQ-1: The system shall collect and summarize data on information, members details, deposits, withdrawals, loans and dividends.

REQ-2: The system shall generate the report in a standardized format for the board director to review.

REQ-3: The system shall save the reports archive for documentation use and notify the board director when it is available by admin.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Performance requirement	Rationale
Registration and Login Process - The system should allow users to choose either register or log in to the system. - The new user registration should be processed within 2 seconds and authenticate logins within 1 second.	- It will provide a clear entry path for both new and old users to access the system. - Fast response time can enhance user experience.
Loan Application Processing - The system should update the new loan application status after the user has taken a new action within 5 seconds.	Users will increase their confidence due to the quick response time.
Notifications and Alerts - Automated notifications such as email or SMS should be sent within 30 seconds of a relevant action done by the users.	Users will be informed by timely notifications of their account status or loan status.
Administrative Data Verification and Updates - Admin and board director should review and do approval or reject action within 2 hours of submission.	In order to ensure that users receive timely updates on their inputs, this requirement is essential for maintaining responsiveness and timely processing in the system.

5.2 Safety Requirements

Safety requirement	Safeguards
Data Protection and Privacy - The system must comply with the Personal Data Protection Act (PDPA) Malaysia and other laws related to data privacy to ensure that users' personal and financial information is stored securely and used only for authorized purposes.	System developers should implement end-to-end encryption for data storage and transfer including personal and financial information.
Access Control and Authentication - Only verified users should have access to the system.	System developers should implement two-factor authentication (2FA) for all users to prevent unauthorized access. Apply role-based access control to ensure users only have access to their relevant information and functions.
Regular Data Backups and Recovery Protocol - The system should be backed up daily automatically to ensure data integrity and	System developers should implement a backup and recovery protocol to restore data within 1 hour of any data loss incident.

recovery in case of any data corruption, cyber-attacks or other system failures.	
User Education and Training - The system should provide all users with training on data security protocols, like safe password management.	The system should prompt a safe password formula to guide users to set a safer password for their account.

5.3 Security Requirements

Security and privacy requirements	Security or privacy certifications
User Identity Authentication - Two-factor authentication (2FA) must be undergone by all users when logging into their accounts.	2FA adds an extra degree of protection to user accounts by preventing unwanted access even in the event that login passwords are stolen.
Role-Based Access Control (RBAC) - Users have restricted access to their user type due to role-based access control.	RBAC protects sensitive data from unauthorized individuals from accessing or changing information beyond their assigned rights.
Data Encryption - All financial and personal information must be encrypted including financial records, loan application information and member data.	This requirement protects data from being intercepted or accessed by unauthorized parties to make sure that all data remains confidential.
Compliance with Personal Data Protection Act (PDPA) - The system must adhere to PDPA Malaysia which requires that personal data be handled, processed and stored responsibly including data privacy information to be given.	The cooperative guarantees the protection of users' privacy rights, safeguarding personal information and minimizing legal risks.

5.4 Software Quality Attributes

Additional quality characteristics	Rationale
Availability - The system must be available for all users at least 99.9% of the time during peak hours and 99.7% overall uptime on a monthly basis.	The users can consistently access the system from time to time for time-sensitive operations like loan applications and approvals.
Usability - User tasks should require no more than	Users' satisfaction will increase when a user-friendly interface is provided.

five steps to complete with clear on-screen instructions and prompts.	
Maintainability - The system's maintenance tasks such as fixing issues should not take longer than two hours of system downtime.	Efficient maintenance practices minimize downtime and guarantee that the system can be readily modified for regulatory updates or feature improvements.
Testability - Each functional requirement should be tested under conditions of simulated peak load. 95% of user actions should be covered by automated testing for common workflows.	The system can manage reasonable user expectations, testability promotes continued reliability and maintainability by spotting any problems at an early stage.

5.5 Business Rules

Individuals	Permission	Safeguards
Member (User) - register, login, apply for loans, check their application status, view their savings, see personal financial statements	Can only see and manage their own information in their account.	Restricted from accessing other users' data.
Admin (User) - Update application statuses and manage users' information	Can access all member data necessary for managing member information.	Time limitations on certain tasks and action logs for accountability.
Board Director (User) - Review and approve or reject member applications and loan applications, oversee the loan process, check system audit logs	Can review all data.	Decisions are logged for transparency.
System Administrator - Handle system settings, maintenance, and security updates	Full control over system configurations but have no access to modify users' data directly.	Limited access to sensitive data with actions logged for accountability.

6. Other Requirements

<Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.>

Appendix A: Glossary

<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.>

Appendix B: Analysis Models

<Optionally, include any pertinent analysis models, such as data flow diagrams, class diagrams, state-transition diagrams, or entity-relationship diagrams>

Appendix C: To Be Determined List

<Collect a numbered list of the TBD (to be determined) references that remain in the SRS so they can be tracked to closure.>

